



An asymmetrical approach to understanding consumer characteristics in banking trust during the COVID-19 pandemic in Italy

Aymen Ammari^a, Evita Allodi^b, Dario Salerno^c, Gian Paolo Stella^{c,*}

^a Department of Finance - INSEEC Business School, 18 Quai de Bacalan cs 9104, 33070 Bordeaux, France

^b Department of Economic and Management - University of Parma, Via J.F. Kennedy, 6, 43121 Parma, Italy

^c Department of Business Economics and Quantitative Studies - University of Naples "Parthenope", Via Generale Parisi, 13, 80132 Naples, Italy

ARTICLE INFO

JEL Classification:

G21
G41

Keywords:

COVID-19
Pandemic
Trust
Risk perception
Banking system
Conspiracy

ABSTRACT

This study aims to examine how perceived banking management, the perception of the context in which the banking system operates, and other consumer characteristics have interacted to create different recipes of banking trust during the second wave which began in October 2020 of the COVID-19 pandemic in Italy. To scrutinize this configurational reasoning, fuzzy set qualitative comparative analysis (fsQCA) was implemented. Using a sample of 1319 Italian respondents, this study suggests that high financial literacy combined with an optimistic attitude during the pandemic are key factors for achieving high banking trust. Furthermore, the results suggest that high perceived conspiracy is linked to low banking trust. These findings can help managers design differentiated strategies to attract depositors and invite the Italian government to double its efforts in restoring the people's optimism.

1. Introduction

Banks play a key role as intermediaries in the economic system between savers and businesses or households. They collect savers' savings and make them available to those who need them in the form of loans. All this is possible only when there is a relationship of trust. At the individual level, as defined by Ennew and Sekhon (2007), a prerequisite for trust in the banking system is interdependence between the customer and the bank: customers and banks must rely on each other, without knowing in advance how the other party will behave.

As a consequence of this interdependence, banks are inherently susceptible to the level of trust individuals place in them. Circumstances and context affect trust (Nooteboom, 2010) through psychological dynamics that cause panic, thus "shifting expectations" of individuals and resulting in bank runs (Guiso, 2010; Dufwenberg, 2015; Dijk, 2017). The bank run phenomenon tends to occur during crises and can further exacerbate them (Goedde-Menke et al., 2014).

On March 11, 2020, the World Health Organization declared a health pandemic caused by the COVID-19 virus. The pandemic led to stock market crashes, spikes in volatility in several stock markets, and, consequently, significant turbulence in global financial markets (Ashraf, 2020; Liu et al., 2021; John and Li, 2021). Italy was the first Western country to have to manage large numbers of people

* Corresponding author.

E-mail addresses: aammari@inseec.com (A. Ammari), evita.allodi@unipr.it (E. Allodi), dario.salerno@uniparthenope.it (D. Salerno), gianpaolo.stella@uniparthenope.it (G.P. Stella).

<https://doi.org/10.1016/j.ribaf.2023.101903>

Received 8 November 2021; Received in revised form 19 January 2023; Accepted 7 February 2023

Available online 10 February 2023

0275-5319/© 2023 Elsevier B.V. All rights reserved.

infected with the virus and therefore to face large-scale health and socioeconomic challenges (Berardi et al., 2020). To slow the spread of COVID-19, the Italian government enforced a sweeping set of nationally homogeneous restrictions requiring social distancing at all levels of society.

During the pandemic, the Italian Banking Association has revealed an arrangement with several professional associations to set up debt repayment by incorporating small loans, mortgages and various lines of credit, while the Development Finance Institution and the National Institute for Promotion have enlarged the funding boundaries for the banking system from €1 million to €3 million. These measures offer sponsored loans to SMEs and support investments within the country. The increased availability of loans and funding are intended to alleviate the perception of possible risk and trust in the financial system by affording liquidity assistance to banks and ensuring that strong deposit insurance systems remain in place to safeguard financial stability.

The current COVID-19 health crisis is far different from the recent and previous economic crisis of 2008–2009. Both crises strained global economies and financial markets, but unlike the 2008–2009 crisis, where the events leading up to the crisis were endogenous, the current health crisis resulted from the new coronavirus and represents a purely exogenous shock to the global financial system (Hasan et al., 2021).

For these reasons, it is important to understand whether and how individuals' perceptions of this health crisis lead to high or low trust in the banking system.

Existing studies have focused mainly on the level of trust in banks in general and during financial crises with reference to the characteristics of individuals, but they have not considered the role played by people's thoughts and perceptions in relation to the context and characteristics of the banking institution. What is missing in the previous literature is a configurational reasoning in which all these variables interact to determine level of trust.

Social reality is characterized by a complexity due to which several combinations of conditions can lead to a particular outcome. This complexity is exceptionally relevant to the context of individual trust since people differ in their profiles, perceptions and demographics, each of which influence their propensity towards trust.

To address the research gap, this study implements a fuzzy set Qualitative Comparative Analysis (fsQCA) on a sample of 1319 Italian respondents during the "second wave" which began in October 2020 of the COVID-19 pandemic in Italy. The results suggest that financial literacy combined with individual perceptions of COVID-19 pandemic are key elements when it comes to trust in financial institutions. In other words, the findings suggest that financial literacy alone does not produce high banking trust and that an optimistic attitude during the pandemic is the missing piece of the puzzle. Furthermore, the results suggest that personal perceptions of the COVID-19 event may have increased or decreased trust in the banking system.

This study makes several contributions. Firstly, we provide new research insights into the recent literature on the effects of the COVID-19 shock for financial markets, particularly concerning trust in the banking system. Secondly, the paper identifies multiple, equally effective combinations of conditions that explain high/low banking trust, which may help banking institutions to better understand certain attitudes or behaviors among account holders. Thirdly, the study applies an fsQCA, which provides a unique middle ground between qualitative and quantitative methods, particularly suited for our purpose.

The remainder of the paper is organized as follows. Chapter 2 presents the theoretical background and literature review. Chapter 3 describes the methodology, the sample selection process, and the approach to measuring the relation between trust in the banking system and the COVID-19 pandemic. In Chapter 4, we outline the results and engage in discussion. We conclude in Chapter 5.

2. Theoretical background and literature review

2.1. Trust

Rousseau et al. (1998) defined trust as "a psychological state comprising the intention to accept vulnerability based upon positive expectations of the intentions or behavior of another" (pp.395). More specifically, trust is the expectation that another person (or institution) will perform actions that are not harmful to us regardless of our ability to monitor such actions (Sapienza and Zingales, 2012).

To schematize this concept, it is possible to state that a high or low level of trust involves a trustee A trusting a trustee B regarding a specific behavior/expectation X in a specific context Y at time t (Uslaner, 2018). Stating that "a trustee A trusts B" presupposes the idea that individuals tend to trust to a greater or lesser degree according to some of their intrinsic characteristics: this is a "basic" level of trust from which situational expectations may deviate in a positive or negative direction (Uslaner, 2018). Trust can also change over time, meaning that a trustee can change their expectations over time, and we can replace these parameters with different trustees, behaviors and real-life contexts.

Therefore, there are three factors that determine trust (De Jager, 2017): (i) the characteristics of the individual who trusts, (ii) the context in which the relationship of trust is established and maintained, and (iii) the characteristics of the subject in whom one believes. Since the definition of trust implies subjective elements of perception that may vary for each individual, the instrument most commonly used to measure trust is the questionnaire. Likert scales in particular are capable of transforming a qualitative measure into an objective one. Based on the World Values Survey, General Social Survey (US), and British Household Panel Study, questions used to measure trust include "On a scale of 1–5, how much trust do you have in 'B'?" or "In general, would you say that most people can be trusted, or that you cannot be too careful in dealing with people?".

The use of the questionnaire as a research instrument allows the researcher to interview numerous people in a relatively short time and to select respondents a priori on the basis of the specific focus of the research. However, attempting to measure trust by means of questions has its limitations; these are generic questions referring to a concept that can be interpreted by the respondent in different

ways in relation to a specific event in the respondent's mind (Ermisch et al., 2009).

2.2. Trust in banking system

Over time, an extensive literature has developed on bank trust (Buriak et al., 2019; Omoregie et al., 2019; Sharma, 2019; Lin et al., 2020; Bilisbekov et al., 2021; Cahyono et al., 2021). In particular, research on trust in banking has mainly focused on two aspects: trust in the individual intermediary to whom the customer entrusts themselves (narrow-scope trust) and trust in the banking system in general, without any specific reference to a single institution (broad-scope trust; Van der Crujisen et al., 2021b). According with Cahyono et al. (2021), using a sample of 217 customers during the COVID-19 pandemic, research demonstrated that bank interest significantly affects depositors' trust. Additionally, the results of cross-sectional estimations have confirmed a positive link between interpersonal trust and trust in banks (Buriak et al., 2019). The cross-sectional analysis of Lin et al. (2020) suggests that companies must strengthen areas such as liquidity monitoring, information security, and compliance with financial regulations to gain customers' trust in the banking sector. This research focuses on the concept of trust in the banking system in general. This kind of trust depends – like the general concept of trust – on three main factors: (i) the characteristics of the individual who trusts, (ii) one's individual perception of the context in which the banking system operates, and (iii) the perceived characteristics of the banking system itself (Fungáčová et al., 2019).

2.2.1. Trust in the banking system: Perceived characteristics of the banking system

When talking about trust, there is also a risk of trust being betrayed (Mayer et al., 1995; De Jager, 2017), and the individual's perception of risk plays an important role in determining their willingness to trust. The relationship between citizens and the system of economic institutions (which comprises the banking system) can be represented within an agent-principal model in which citizens are the principal and economic institutions the agent. When the principal for whatever reason loses the trust initially placed in the agent, it is possible that the citizen will want a change and thus a new agent; in the worst case scenario for the banking system, this loss of trust may lead to a bank run (Wälti, 2012). Consequently, it is extremely important for the banking system to continue to perform its function by benefiting from the trust of citizens, without whose savings it would not only fail to survive, but it would also damage the economy by reducing the availability of capital for productive purposes (Jaffer et al., 2014). This trust must be established on the basis of factors/characteristics of the system that can be observed by citizens.

The perceived characteristics of the banking system are substantiated by the belief that the institutions that jointly constitute the banking system are sincere, benevolent, trustworthy, transparent and honest and that they observe rules and pursue general interests (Devos et al., 2002; Grayson et al., 2008; Hansen, 2012; Jarvinen, 2014). Carbó-Valverde, Maqui-López, and Rodríguez-Fernandez (2013), through the analysis of a dataset submitted to Spanish bank customers (N = 1600), analyzed the perception of certain banking characteristics such as effectiveness, sensitivity to customer issues, and commitment and estimate the effect these characteristics have on the banking system in general. The researchers found that perceptions that banking institutions are attentive to the needs of their customers and are committed to researching and proposing suitable instruments for specific needs positively influence the trust that customers have in the individual institution and in the banking system as a whole.

When a customer perceives non-transparent or ambiguous behavior on the part of the banking system, they may develop conspiracy theories according to which people/institutions/banks collude to achieve a hidden goal that the citizen perceives as illegal or malicious (Van Prooijen and Acker, 2015). In a context of uncertainty like that which resulted from the COVID-19 health crisis, it is possible for individuals to develop conspiracy theories and doubt the honesty of others, especially when it comes to banking institutions that were able to manage the COVID-19 pandemic as a threat and transform it into a growth opportunity that improved their performance. Indeed, Li et al. (2021) used a sample of 215 US banks to examine the relationship between the use of non-interest income sources and banks' earnings and risk in the context of the economic crisis caused by the COVID-19 pandemic. Banks, following the tightening of lending conditions, reorganized their assets by opting for greater diversification. Li et al. (2021) identified a positive relationship between performance, as measured by return on assets and return on equity, and non-interest income.

2.2.2. Trust in the banking system: characteristics of the individual who trusts

The characteristics of the individual were analyzed with the first studies on the concept of trust in general, which did not feature a specific link to the banking system (Knack and Keefer, 1997; Alesina and La Ferrara, 2002; Bjørnskov, 2007). Low education, low income level, and the presence of psychological trauma are indicators of the profile of a person with low trust.

In a cross-country study, Fungáčová et al. (2019) used banking trust data from the World Values Survey (WVS) for the years 2010–2014 for 52 countries and determined that trust in the banking system is positively correlated with income and negatively correlated with level of education. With reference to income, it is reasonable to assume that individuals with higher incomes tend to have more trust in the banking system as a result of more frequent interactions with the banking system.

The variable of education refers to the level of education attained by the respondent. This variable can be further narrowed to refer to the financial knowledge of individuals and thus their degree of financial literacy. Recently, van der Crujisen et al. (2021) used a regression model based on an analysis of 14 years of data on Dutch consumers' trust in financial institutions to determine that financially literate consumers are more likely to trust the banking system. Financial literacy is necessary both for a bank-customer relationship in which the latter has sufficient knowledge to assess the characteristics of the institution and when knowledge is absent or insufficient, which means one must blindly trust the banking system (Gaspar et al., 2020).

In order to strengthen the link between trust and financial system stability, the literature provides several pieces of evidence regarding the role of financial literacy in determining individuals' levels of trust in the financial system (see Hansen, 2012, 2014; Shim

et al., 2013; Lusardi and Mitchell, 2014; Kersting et al., 2015; van der Crujisen and Jonker, 2019; van Der Crujisen et al., 2021a; Fisch and Seligman, 2022). In particular, according to Letamendia and Poher (2020), individuals who exhibit lower levels of financial literacy are those who demonstrate less trust in banks and the financial system. In contrast, Kersting et al. (2015) found that a higher degree of financial literacy is associated with lower levels of trust in the financial system.

Finally, personal experiences also have an impact on one's level of trust in the banking system. Through a dataset created by merging Laeven and Valencia's (2018) dataset on the subprime mortgage crisis and the 6th wave of the World Value Survey, which contains information on trust in banks and individual characteristics for 52 countries, Fungacova et al. (2022) indicated that the experience of a banking crisis decreases trust in banks. Notably, when a banking crisis is severe, there is a decrease in trust that lasts for a long time among individuals who are young at the time of the crisis. In contrast, when the crisis is not severe, the greatest impact is on adult individuals. This result aligns with findings in the literature on the effects of crises on economic beliefs and behavior, according to which early life crisis experiences can exert long-term effects on beliefs (Malmendier and Nagel, 2016; Giuliano and Spilimbergo, 2014).

2.2.3. Trust in the banking system: the context

Following the financial crisis of 2008–2009, researchers began to study the phenomenon of trust in relation to banking institutions, shifting their research from the characteristics of individuals to the context of the crisis. Sapienza and Zingales (2012) examined the level of trust in the financial system among US households ($N = 1034$) in 2008. Respondents were asked to quantify their level of trust in various institutions on a scale of 1–5. According to this research, banks are perceived as more trustworthy than the stock market and the government but less trustworthy than people in general.

To analyze the conditions under which people might lose trust in banks, Jansen et al. (2015) used 2500 Dutch respondents to investigate possible scenarios associated with the financial crisis. These scenarios refer to negative news, government policies, and lack of transparency. Respondents were asked whether these possible scenarios could erode their trust in the banking system. Jansen et al. observed that confidence in banks is strongly influenced by these scenarios. In particular, negative media reports, possible remuneration bonuses for bankers, and opaque information about the products offered are among the causes that most reduce the level of trust in banks.

Knell and Stix (2015) administered a quarterly questionnaire ($N = 2000$) from 2004 to 2013 in Australia and highlighted that levels of trust in the banking system are related to individuals' subjective views of the economic situation, the perceived and expected performances of banks, and individuals' direct experiences of bank failures. Unfortunately, there is only one study on how confidence in the banking system is affected by a non-financial crisis like the COVID-19 pandemic.

Van der Crujisen et al. (2022) conducted a survey in 2021 to examine the drivers of trust in banks in the United States ($N = 2076$) and the Netherlands ($N = 2563$) and if COVID-19 may have influenced consumer trust. Their analysis showed that the COVID-19 pandemic had no particular effect on trust in financial institutions, although the results differed between the two countries considered. In particular, the authors found that individuals in poor health had lower levels of trust than healthy people and that trust among respondents in poor health declined more during the pandemic (Van der Crujisen et al., 2022). As indicated by Van der Crujisen et al. (2022), this result – in contrast to previous findings regarding the effects of the economic crisis on people's level of confidence – requires further investigation. The outcome may have been influenced by the period in which the survey was conducted, respondents' direct experiences with the virus, and certain benefits provided by credit institutions during the various lockdown periods.

However, given that previous studies suggest that circumstances and context can influence individuals' confidence by acting on

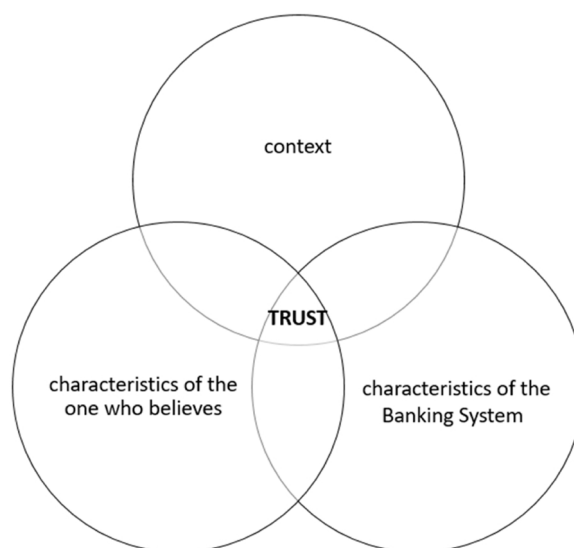


Fig. 1. Conceptual Model.

psychological factors, it is worth noting that recent literature regarding the COVID-19 pandemic identifies a negative impact of virus containment measures that fostered excessive psychological distress (Flesia et al., 2020; Liu et al., 2020; Losada-Baltar et al., 2020), forcing individuals to seek new ways to cope with and adapt to the emergency (Orrù et al., 2020; Polizzi et al., 2020; Wang et al., 2020; Patel et al., 2022). In addition, many conspiratorial beliefs have appeared along with the COVID-19 pandemic (Bertin et al., 2020; Georgiou et al., 2020) that have been associated with undesirable attitudes towards trust.

2.3. Conceptual model

Although the issue of trust within the banking system has been the subject of more studies in recent years than in the past, what has not yet been studied is a configurational reasoning (Fig. 1 – Conceptual Model) in which the variables belonging to the three aforementioned factors – namely (i) the characteristics of the individual who trusts, (ii) one's individual perception of the context in which the banking system operates, and (iii) the perceived characteristics of the banking system itself – interact together to determine a high (low) level of trust.

In contrast to previous studies in which the methodology is quantitative and reliant on the use of regression models, the fsQCA approach leads to a more comprehensive understanding of the conditions under which a high (low) level of trust occurs. Whereas regressions explain the effects of particular causes, fsQCA makes it possible to identify the minimally necessary and/or minimally sufficient conditions (combinations of) that lead to an effect. This technique is exceptionally relevant to the context of individual trust since people differ in their profiles, perceptions and demographics, each of which influence their propensity towards trust.

The aim of this article is not to estimate the average effect of one (or more) variables in a population of cases (i.e., the effects of causes approach) as in previous studies but rather to identify the minimum conditions (combinations of) that lead to a particular outcome in specific cases (i.e., the causes of effects approach). Furthermore, the model in the present research considers the COVID-19 variable that assumes the role of a contextual causal factor, but at the same time it allows the interaction with the characteristics of the banking system that are perceived by those who choose to trust.

Thus, the research question consists in identifying the minimum and/or necessary causes for a high (low) level of trust in the banking system in the economic context of a health crisis within a configurational approach.

Table 1
Descriptive statistics (N = 1319).

Variables		Percentage %	National Statistics %
Age	15–24 years old	12.9	9.8
	25–49 years old	59.7	31.2
	50–64 years old	20.9	22.6
	65–79 years old	6.4	16.0
	> 80 years old	0.2	7.6
Gender	Female	42.8	52.7
	Male	57.2	48.3
Marital Status	Married	39.7	47.4
	Single/Unmarried	53.5	42.3
	Divorced	5.4	7.3
	Widowed	1.4	3.0
Education	< 0–9 Years	10.5	16.2
	9–12 Years	17.5	32.4
	> 12 Years	72.0	51.4
Work Status	Full-time work	49.0	37.4
	Part-time work	13.4	9.0
	Stay-at-home	18.4	17.9
	Students	5.5	11.3
	Retired	7.3	18.3
	Disabled or unable to work	0.2	1.2
	Unemployed	6.2	4.9
Income	Less than \$10,000	11.8	12.3
	€10,000–€20,000	21.5	24.9
	€20,000–€40,000	32.1	38.3
	€40,000–€80,000	12.6	18.8
	€80,000 or more	3.9	5.7
	Prefer not to say	18.2	/
Stock Market Participation	No	75.0	66.0
	Yes	25.0	34.0

Note: National statistics were provided by both the Italian National Institute of Statistics and Eurostat.

3. Methodology

3.1. Survey design and sample features

To pursue the objective of this research, we launched an online survey to collect data related to whether respondents invest in financial markets, their level of financial literacy, perceived banking management effects, and expectations arising from the pandemic.

We collected data by administering questionnaires in Italy from the beginning of November 2020 to the end of December 2020 via the Computer-Assisted Web Method (CAWI). The questionnaire was promoted on social media platforms such as Facebook, Twitter, and LinkedIn. Our sample appears to be representative of the Italian population following the indications provided by [Dattalo \(2008\)](#).

We invited 2000 randomly selected individuals and, after cleaning the sample and eliminating questionnaires with missing answers, we ended with 1319 respondents (response rate = 65.9 %). Almost half of our respondents (35.3 %) were in the age range 26–35 %, and 14.3 % were 36–45. Only 5.5 % were students, while almost a half of respondents (49.0 %) were full-time workers and 13.4 % worked part-time. Males represented 57.2 % of our respondents, and 39.7 % were married. The remaining respondents were single (either unmarried, divorced, or widowed). In terms of education, 72 % of respondents had more than 12 years of schooling, 17.5 % had completed between nine and 12 years, and 1.4 % had completed less than nine years. In terms of income, approximately half of the respondents earned 20,000–40,000 euros yearly. Regarding stock market participation, only 25.0 % of the sample had invested in financial markets (see [Table 1](#)).

3.2. Measures

To measure if respondents had invested in financial markets, we used the approach proposed by van Rooij et al. (2011). To quantify financial literacy, we used a methodology proposed by Standard & Poor's Rating Services Global Financial Literacy Survey (S&P Global FinLit Survey; [Klapper and Lusardi, 2019](#)). The outcome of each question determined the respondent's knowledge of a specific financial theme. There was only one correct option for each question. The dependent variable that is the subject of this study is trust in the banking system. The methodology used was taken from the work conducted by [Sapienza and Zingales \(2012\)](#). To measure the impact of COVID-19 on trust in the banking system, we proposed three items: perceived profit gained from COVID-19, perceived conspiracy, and perceived effective COVID-19 management. These items were measured on a scale from 1 to 7, with 1 identifying low levels of trust and 7 identifying high level of trust in banks. Finally, we asked respondents to indicate whether COVID-19 caused a change in their wealth on a scale of 1–7 (1 = strongly disagree and 7 = strongly agree; see Appendix 1).

3.3. Research model: fsQCA

According to [Ragin \(2000\)](#), the primary assumption of QCA is configurational reasoning. Configurational reasoning states that real phenomena are illustrated by several interactions among real attributes. Rarely does an outcome have a single cause; instead, it is the product of the interdependence of several factors. The main benefit gained from fsQCA is that it supports "equifinality." Equifinality is also called "multiple causations," where "multiple" implies the number of different paths (also known as "recipes") that result in the same outcome (see [Fig. 2](#)). In other words, high levels of interaction among our conditions could define multiple routes to the same level of trust.

In addition, fsQCA supports the incidence of "conjunctural causation"; this means that, for an outcome to occur, the presence and absence of a causal condition depend on how this causal condition combines with one or more other causal conditions. As shown in [Fig. 3](#), QCA is based on the principles of configuration theory, which fundamentally permits the inspection of interplays that develop

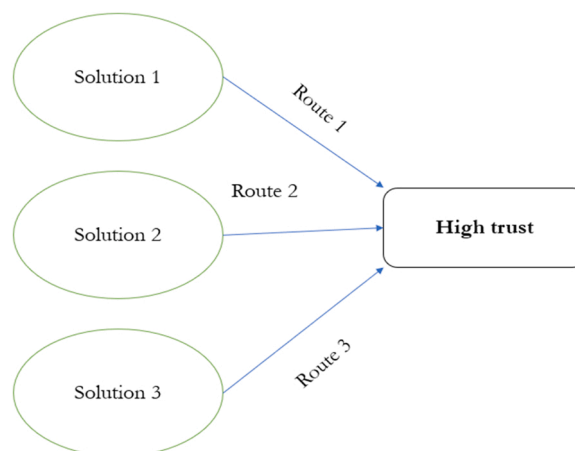


Fig. 2. Equifinality.

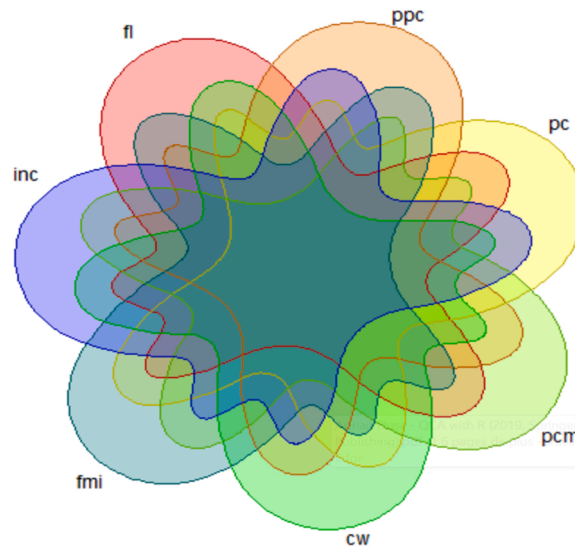


Fig. 3. Conjunctural causation of the seven conditions. Note: (fmi): investing in the financial market, (fl): financial literacy, (ppc): perceived profit gained from COVID-19, (pc): perceived conspiracy, (pcm): perceived effective COVID-19 management, (cw): COVID-19 impact on wealth, and (inc): income.

between different conditions. More specifically, the model of the present study allows for the interaction between (i) investing in the financial market (fmi), (ii) financial literacy (fl), (iii) perceived profit gained from COVID-19 (ppc), (iv) perceived conspiracy (pc), (v) perceived effective COVID-19 management (pcm), (vi) COVID-19 impact on wealth (cw), and (vii) income (inc).

A strong argument for the use of fsQCA can be made when causality is both multiple and conjunctive. When causality is multiple (see Fig. 2), the outcome (bank trust) has more than one cause. When causality is conjunctive (see Fig. 3), the causes of financial literacy, individual perceptions of the COVID-19 pandemic, and individual characteristics interact with each other to produce the outcome. Again, this technique is particularly relevant to the context of customers' behavior in the banking industry because customers differ in the aspects that trigger their trust.

Thus, fsQCA seems to be the long-awaited response to the query of how to detect different configurations (e.g., Harms et al., 2007) of multiple interrelated variables that all lead to the same desired output (in this case, bank trust).

Set-theoretic relationships are, to a large extent, asymmetrical. This asymmetric reasoning is one of the major attributes of QCA that distinguishes it from mainstream statistical methods, which are typically symmetrical by design. For instance, in our sample, there is evidence of asymmetry. As is indicated in Fig. 4, many respondents who did not perceive effective COVID-19 management by the financial institution may still intend to trust it, while some respondents who perceived effective management of COVID-19 by the financial institution were unwilling to trust it regardless. Consequently, a systematic explanation of the mechanism underlying these

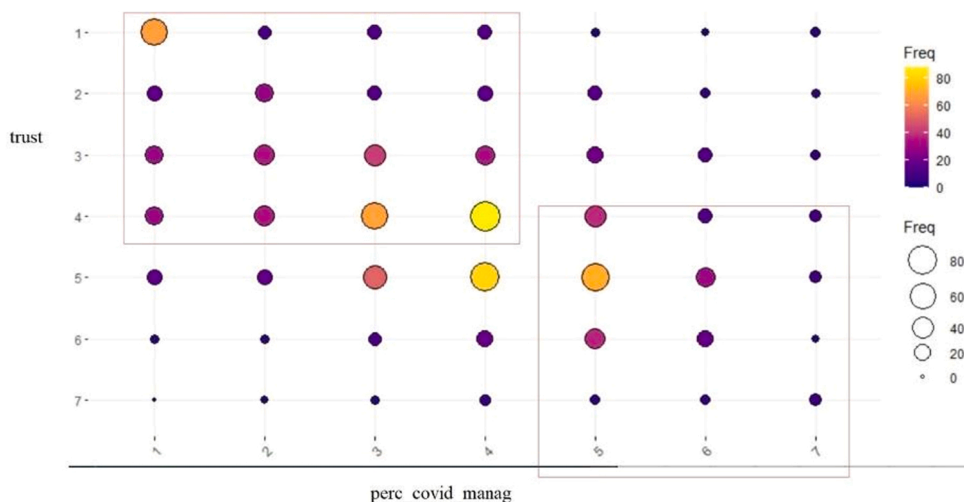


Fig. 4. Asymmetric relationships.

Table 2
Calibration.

	<i>Likert Score = 1</i>		<i>Likert Score = 2</i>		<i>Likert Score = 3</i>		<i>Likert Score = 4</i>		<i>Likert Score = 5</i>		<i>Likert Score = 6</i>		<i>Likert Score = 7</i>	
	<i>Un-calib.dat</i>	<i>Calib. Data</i>	<i>Un-calib.dat</i>	<i>Calib. Data</i>	<i>Un-calib.dat</i>	<i>Calib. Data</i>	<i>Un-calib.dat</i>	<i>Calib. Data</i>	<i>Un-calib.dat</i>	<i>Calib. Data</i>	<i>Un-calib.dat</i>	<i>Calib. Data</i>	<i>Un-calib.dat</i>	<i>Calib. Data</i>
Conditional variables														
Financial literacy (fl)	1	0	2	0.2	3	0.3	4	0.4	5	0.7	6	0.8	7	1
Perceived profit gained from COVID-19 (ppc)	1	0	2	0.2	3	0.3	4	0.4	5	0.7	6	0.8	7	1
Perceived conspiracy (pc)	1	0	2	0.2	3	0.3	4	0.4	5	0.7	6	0.8	7	1
Perceived effective COVID-19 management (pcm)	1	0	2	0.2	3	0.3	4	0.4	5	0.7	6	0.8	7	1
COVID-19 impact on wealth (cw)	1	0	2	0.2	3	0.3	4	0.4	5	0.7	6	0.8	7	1
Income (inc)	1		2	0.2	3	0.4	4	0.7	5	1				
Investing in the financial market (fmi)	0		-		-		-		-		-		1	1
Outcome variables														
Trust on Banking System (trust)	1	0	2	0.2	3	0.4	4	0.4	5	0.7	6	0.8	7	

asymmetric relationships may offer important insights.

3.4. fsQCA procedure

The initial step in the analysis is calibrating dependent (the outcome) and independent variables (the conditions) into fuzzy or crisp sets. Fuzzy sets values range from 0, which denotes an absence of set membership, to 1, which indicates full set membership. Crisp sets are more fitting with binary conditions that have two values (0 or 1).

QCA starts by listing and counting all cases. In our study, we had seven conditions, {fl,ppc,pc,pcm,cw,inc,fmi}, and thus there were ($2^7 = 128$) possible types of observations determined by the possible combinations of conditions, not all of which would necessarily occur. By counting the number of observations that exist for each of the 128 unique combinations of conditions, QCA can identify which implications or descriptive inferences are empirically supported (identifying the necessary and/or sufficient conditions for producing an outcome).

In the next step of QCA, Boolean algebra is used to display the simplest solutions, which reduces the number of inferences to the minimum set of inferences from a truth table. This reduced set of inferences is also called the logical, or Boolean minimization, and it uses "prime implicates." For example, with any two given expressions that differ by exactly one literal, that literal can be minimized.

The QCA method is completely quantitative in determining the results. The technique uses the laws of Boolean algebra to calculate the membership score of each condition and/or configuration of conditions. Moreover, fsQCA employs the Quine–McCluskey algorithm (QMC) to obtain the different solutions.

4. Results and discussion

4.1. Calibration

To calibrate data, we used adjectival phrases to differentiate between a variable and a set. For example, "trust" is a variable, while "high trust" is a set; "financial literacy" is a variable, while "high financial literacy" is a condition. In these examples, only the adjectival phrase refers to a set in which observations may have a membership. Next, we established a careful reflection upon the nature of each measure and its meaning to ensure a successful calibration. In some cases, it was sufficient to use crisp (0–1) sets to represent a variable: for example, the variable "investing in financial markets," when translated into a set, automatically becomes crisp as it can take only two values. For more non-binary variables, the fundamental recommendation is to identify three values from the range of the variables that are to correspond to full non-membership, the most ambiguous membership (cross-over point), and full membership. The most straightforward procedure is to use the three values [0,0.5, 1] and to convert other values based on a linear function to fit into these three values if there is insufficient knowledge to calibrate the variables. Instead, we investigated what each item meant substantively in order to align with substantive and theoretical knowledge. Specifically, we used qualitative anchoring to establish a connection between a fuzzy membership function and the original data.

For the binary variable "investing in financial markets," we used membership values of 0 and 1. For other conditions and the outcome variable, measured on a 7-item Likert scale, the two extreme items were converted to the membership values of 0 and 1, and the intermediate items were assigned membership values in a sub-linear way, being given values of .2,.3,.4,.6, and .8 instead of the equidistant choices of .166,.33,.5,.66, and .82. This meant that a higher score on the Likert scale was mandatory from the respondent for intermediate memberships and revealed the overall insight that the points on the Likert scale were not equidistant. Furthermore, using this substantive knowledge instead of applying statistical distribution of the variable (the mean/median) to

Table 3
Summary of necessary conditions for High/Low Trust.

Conditions	High Trust			Low Trust		
	inclN	RoN	covN	inclN	RoN	covN
Financial literacy (fl)	0.455	0.949	0.836	0.366	0.943	0.816
Perceived profit gained from COVID-19 (ppc)	0.771	0.64	0.597	0.786	0.733	0.739
Perceived conspiracy (pc)	0.272	0.91	0.61	0.307	0.96	0.836
Perceived effective COVID-19 management (pcm)	0.714	0.875	0.793	0.525	0.832	0.706
COVID-19's impact on wealth (cw)	0.875	0.479	0.557	0.843	0.538	0.65
Income (inc)	0.61	0.876	0.754	0.516	0.885	0.775
Investing in financial market (fmi)	0.28	0.836	0.469	0.261	0.852	0.531
Exploring Possible Necessity Relations						
~ppc+pcm+inc	0.907	0.622	0.648			
fl+ ~ppc+pcm+ ~cw	0.907	0.617	0.645			
~ppc+pc+ ~pcm				0.917	0.606	0.719
~ppc+ ~pcm+ ~cw				0.911	0.604	0.716
pc+ ~pcm+ ~cw				0.91	0.66	0.745
pc+ ~pcm+inc				0.915	0.628	0.73
~pcm+ ~cw+inc				0.919	0.617	0.726

Notes: Lowercase code indicates the presence of the conditions in the model predicting the outcome. ~ stands for the absence or negation of the conditions.

calibrate the seven-item scale makes it possible to develop a meaningful calibration.

Converting the middle point (neutral value) of a seven-item scale into .5 would mean that a neutral opinion positively contributes to the evaluation of a configuration. For example, for the trust scale, the question was "On a scale ranging from Not at all = 1 to Very much = 7, how much confidence do you have in the banking system?" A respondent who assigned 4 points to this question (as a neutral response) did not imply the presence of trust. To avoid this problem, we chose to use 0.4 as a crossover point.

Similarly, we attributed values of .2, 0.4, and .7 for the variable of income, which was measured on a 5-item Likert scale, instead of the equidistant choices of .25, 0.5, and .75.

Table 2 and shows both the non-calibrated data and the data after the applied calibration criteria.

4.2. Analysis of necessity

A condition is necessary for an outcome if the condition is always present when the outcome occurs and if the outcome does not occur in the absence of this condition. The term "inclusion" (inclN) means the proportion of the set of the outcome that is included in the set of the condition. By convention, a condition is called "necessary" if the inclusion (consistency) score exceeds the threshold of .80 (Ragin, 2009; Wagemanna and Schneiderb, 2010). Fuzzy coverage (covN) is about the proportion of the condition that is covered by the outcome. We added another parameter of fit called RoN (Relevance of Necessity). A necessary condition is trivial (not relevant) if and only if there is no empirical evidence of the negation of the condition. The lower the RoN score, the more trivial a condition is, and the higher the RoN score, the higher the relevance and the greater the relative importance of that condition as a necessary condition.

Table 3 presents the results of the necessary analysis for high and low trust. Furthermore, it includes an overview of all possible necessity relations, eliminating useless tests on expressions that are not necessary for the outcome and specifying a cut-off for the relevance of necessity (at least .6). As the table displays, the impact of COVID-19 on wealth (cw) achieved consistency scores higher than .80 for high trust and low trust, so this condition can be considered a necessary condition, or a prerequisite for high/low trust.

The perception of the impact of COVID-19 on wealth is basically linked to emotions. Emotions can distort and influence decision-making. Fear has particularly significant effects on risk perception (Lerner and Keltner, 2001; Lerner et al., 2004). Moreover, fear, along with anxiety and panic attacks, is an emotion that plays an important role during a crisis (Dijk, 2017) and reduces trust (Myers and Tingley, 2011). Individuals intuitively perceive the COVID-19 virus as a danger that people are or may be exposed to and that is therefore frightening (Pakpour et al., 2020). Furthermore, the information that individuals receive from the mass media every day covering COVID-19 news affects their emotions.

Necessity analysis reveals that high income, positive perception of COVID-19 management, and negative perception of profit gained from COVID-19 are considered as necessary conditions for high trust. From these results, it can be concluded that different factors are necessary for achieving high trust. According to Table 2, different factors are similarly necessary for low trust. In sum, achieving high or low trust requires fulfilling certain criteria.

4.3. Analysis of sufficiency

The analysis of sufficiency relates to the main objective of the QCA procedure to find the minimal configurations of conditions that are sufficient for a given outcome. A condition is sufficient for an outcome when every time the condition is present, the outcome is also present, and when the condition does not occur in the absence of the outcome. While QCA offers the possibility of checking the sufficiency of each condition for the outcome, in this study we performed the sufficiency of each configuration which schematize the interaction between multiple conditions (see Table 4).

Table 4
Sufficiency conditions for high/low trust.

Conditions	High Trust		Low Trust		
	Solution 1.(a)	Solution 2.(a)	Solution 1.(b)	Solution 2.(b)	Solution 3.(b)
Financial literacy (fl)	①	①	①	⊗	⊗
Perceived profit gained from COVID-19 (ppc)	⊗	⊗	①	⊗	⊗
Perceived conspiracy (pc)	⊗	⊗	①	①	①
Perceived effective COVID-19 management (pcm)	①	①	⊗	⊗	⊗
COVID-19's impact on wealth (cw)	⊗	①	⊗	⊗	⊗
Income (inc)		①	⊗	①	①
Investing in financial markets (fmi)	⊗				⊗
Consistency	.982	.983	.972	.972	.97
PRI	.776	.863	.65	.81	.676
Raw coverage	.122	.247	.081	.063	.06
Unique coverage	.025	.15	.007	.019	.003
Overall solution consistency	.981		.967		
Overall solution PRI	.857		.772		
Overall solution coverage	.272		.103		

Notes: ① Presence of the conditions in the model predicting the outcome ⊗ Absence or negation of the conditions. Blank spaces refer to a "not important" condition. Large circles indicate the core conditions, and small circles indicate peripheral conditions. Core conditions are derived from parsimonious solutions and peripheral conditions are from intermediate solutions. Intermediate solutions constitute subsets of the most parsimonious (or simplest) solution.

To identify a sufficient combination of conditions, the first step is the construction of the truth table. The truth table lists and sorts all the possible combinations of conditions and the empirical outcome associated with each configuration. The table contains 2k rows, where k refers to the number of conditions and 2 refers to the two qualitative anchors of low/high or presence/absence for binary variables. With the seven conditions used in our study, there are 128 (i.e., 2⁷) corners to the vector space. These 128 corners match 128 logically possible configurations that can be composed using our conditions. For each of the 1080 cases, row data with missing observations was dropped.

There are relevant configurations (i.e., respondents with greater than .5 membership) of the 128 logically possible combinations of conditions. The other instances were defined as logical remainders. We selected a frequency cut-off of a minimum of 1 case with greater than .5 membership in a combination to detect these relevant configurations.

The next task is to list these relevant combinations according to their consistency scores. The value of this score represents the proportion of cases in the combination that displays the outcome in question. The consistency value was set at .9; this means that combinations with a consistency score equal to or above .9 are labeled 1; otherwise, they are labeled 0. A visual inspection of the truth table is given in Fig. 5.

The most important feature of this figure is its ability to recognize the truth table object and plot all of its intersections according to the output value. The colors are associated with the positive output configurations (green), negative output configurations (orange), contradictions (blue), and the rest of the intersections with no empirical evidence (remainders, white).

The QCA proposes three different solutions by applying the Boolean minimization process: (1) a "complex" solution, (2) a "parsimonious" solution, and (3) an "intermediate" solution.

The first solution does not contain the remainders, while the second and third ones do. The parsimonious solution includes all the remainders, and the intermediate solution contains only those which are plausible for the analysis. We display the intermediate solutions because this solution does not allow the removal of all necessary conditions and is therefore superior to the other solutions (Rihoux and Ragin, 2008).

To assess sufficiency, we used the inclusion (consistency) score, which means the proportion of the set of the condition that is included in the set of the outcome. A condition is deemed "sufficient" if the inclusion (consistency) score exceeds the threshold of .80 (Ragin, 2008; Schneider et al., 2010). Coverage indicates how much of the outcome is covered (or explained) by each solution term and by the solution as a whole. We used the Proportional Reduction in Inconsistency (PRI) score to resolve a situation with simultaneous subset relations in which a condition appeared to be sufficient for both the outcome and its negation (high trust and low trust), which is a logical contradiction. The PRI score was used to declare the condition as sufficient for either the outcome or its negation, as logically a condition cannot be sufficient for both the outcome and its negation. Finally, unique coverage means the proportion of memberships in the outcome explained solely by each solution term (memberships that are not covered by other solution terms).

Table 4 suggests that two configurations are sufficiently linked to high trust and three configurations are sufficiently linked to low trust. Columns 2–6 present the casual paths for high trust (Solution 1. [a] or solution 2. [a]) and for low trust (Solution 1. [b], solution 2. [b] or solution 3. [b]). Ragin (2008) advocated for a solution to be considered informative when the overall solution consistency is above .8; this means that the conditions present in these configurations are highly consistent subsets of the solution. The overall solution coverage indicates the extent to which this condition covers the outcome. Configurations that are sufficiently linked to high trust represent 98.1 % of the total cases and covers 27.2 % of the cases. Solution 1. (a) sufficiently accounts for high trust in 98.2 % of the total cases and covers 12.2 % of all cases. It combines high financial literacy, high perceived effective COVID-19 management, and low

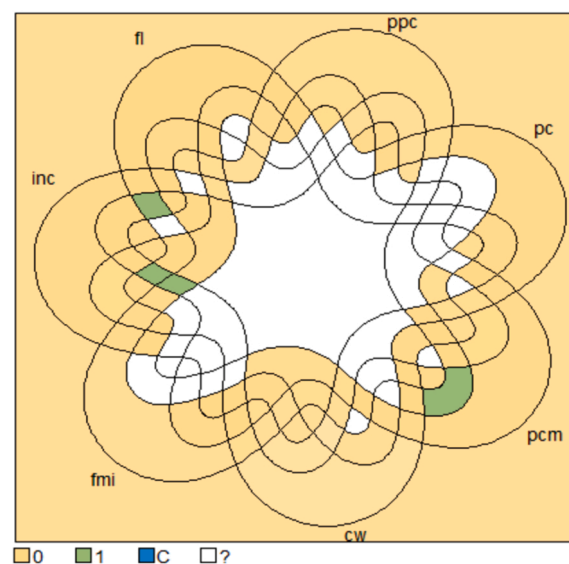


Fig. 5. Visual Inspection of the Truth Table.

perception of profit gained by banks during the pandemic (core conditions) with low perception of a possible conspiracy. This condition leads to high trust in financial institutions for respondents who are not involved in financial market investment and with low concerns about the economic consequences of the COVID-19 pandemic. Solution 2. (a) sufficiently accounts for high trust among 98.3 % of the total cases and covers 24.7 % of all cases. This solution indicates that the same configuration (high financial literacy, high perceived effective COVID-19 management, and low perception of profit gained by banks during the pandemic (core conditions) with low perception of a possible conspiracy) holds and leads to high trust for wealthy respondents who have high concerns about the economic consequences of the COVID-19 pandemic. In this combination, high trust can be achieved regardless of the condition of investment in the financial market, as indicated by the blank space that signals a "don't care" situation. From the two previous solutions (solution 1. [a] and solution 1. [b]), we can identify two different profiles of respondents: (1) respondents who are not involved in financial market investment and who have low concerns about the economic consequences of the COVID-19 pandemic and (2) wealthy respondents who have high concerns about the economic consequences of the COVID-19 pandemic. Interestingly, the two profiles took the same road (the same path of configuration), which emphasizes the centrality of high financial literacy combined with a rosy outlook (high effective management and low conspiracy beliefs) to achieve high banking trust.

These findings are in line with previous research and indicate that combining financial literacy with an optimistic attitude during the COVID-19 pandemic is a guarantee for bank trust. This high level of financial knowledge enables optimistic respondents to correctly identify and delineate the health risk caused by the pandemic and to distinguish it from financial risk. As reported by [Caccagno and Monticone \(2015\)](#), financially literate individuals can search for, understand, and evaluate economic and financial information. The analysis of this information helps them to place their trust in financial institutions, which is not the case for individuals with little financial knowledge. Additionally, in this context of uncertainty and anxiety, this rosy outlook is the missing piece of the puzzle; without this optimism, a person with high financial literacy may not have high banking trust. This finding offers plausible answers to the contradictions, gaps, and inconsistencies in the literature and suggests an approach that goes beyond the simple linear positive or negative relationship between financial literacy and banking trust. At the same time, the asymmetric reasoning of QCA helps to explore the alternative combination for achieving low trust in financial institutions, which is also key to our research. Configurations that are sufficiently linked to high trust represent 96.7 % of the total cases and covers 10.3 % of all cases.

Solution 1. (b) sufficiently accounts for high trust in 97.2 % of the total cases and covers 8.1 % of cases. The solution indicates that for respondents with low income and high financial literacy and who have low concerns about the economic consequences of the COVID-19 pandemic, high perception toward a possible conspiracy (core condition), low perceived effective COVID-19 management, and high perception of profit gained by banks during the COVID-19 pandemic lead to low trust in financial institutions.

Solution 2. (b) sufficiently accounted for high trust in 97.2 % of the total cases and covers 6.3 % of cases. The solution indicates that for respondents with high income and low financial literacy and who have low concerns about the economic consequences of the COVID-19 pandemic, high perception toward a possible conspiracy (core condition), low perceived effective COVID-19 management, and low perception of profit gained by banks during the COVID-19 pandemic lead to low trust in financial institutions.

Solution 3. (b) sufficiently accounted for high trust in 97 % of the total cases and covers 6 % of cases. The solution shows that for respondents who are not involved in financial market investment and who have high income, low financial literacy, and low concerns about the economic consequences of the COVID-19 pandemic, high perception toward a possible conspiracy (core condition) and low perception of profit gained by banks during the COVID-19 pandemic lead to low trust in financial institutions.

Although the profiles of the respondents are different, what the three solutions have in common – and what leads to a low level of trust in the banking system – is the perception of a possible conspiracy. Believing in conspiracy theories is known to undermine prosocial behavior and trust in authorities ([Imhoff et al., 2018](#); [Jolley and Douglas, 2014](#)). One possible explanation for the willingness to believe in a possible secret organization plotting against millions of Italians can be found in individuals' tendency to be optimistic or pessimistic. In fact, according to [Furnham \(2013\)](#), pessimists are more prone to conspiracy theories. It is likely that during periods of high stress, a pessimistic person attempts to explain the events that take place by becoming convinced of the existence of organizations plotting against them.

In sum, our findings support the existence of a joint causality relationship between financial literacy, individual perceptions of the COVID-19 pandemic, and high/low trust in financial institutions. Furthermore, we found that financial literacy is a necessary but non-sufficient condition for achieving high trust. Such evidence suggests that financial literacy on its own cannot produce high banking trust. Our results also indicate that a high perception of a possible conspiracy can reduce banking trust (see [Table 4](#)).

4.4. Robustness checks

In the previous analysis, we presented intermediate solutions because this intermediate solution does not allow the removal of all necessary conditions and is therefore superior to the other solutions ([Rihoux and Ragin, 2009](#)).

However, when a condition (or a phenomenon) is necessary for an outcome, its negation cannot be simultaneously sufficient for the same outcome; therefore, any counterfactual solution containing the negation of a necessary condition should be eliminated from the minimization process when performing the intermediate solution. This process is called the enhanced intermediate solution (EIS). In the EIS, solutions could be obtained given the expectations (allowing for the presence of all necessary conditions) and after excluding the contradictory simplifying assumptions (CSAs).

To check the robustness of our results, we performed a second analysis that identifies these CSAs and excludes them from the minimization process when performing the intermediate solution. The findings from the EIS are equivalent to our previous analysis as the truth table did not identify any contradictions; this finding can be seen in [Figure 6](#).

Furthermore, the results of the proposed QCA approach are compared with the results derived from the multiple regression analysis

given in Table 5.

Financial literacy (fl), Perceived profit gained from COVID-19 (ppc), Perceived conspiracy (pc), Perceived effective COVID-19 management (pcm), COVID-19 impact on wealth (cw), Income (inc), Investing in financial markets (fmi).

The linear regression results support a significant relationship between the different conditions and the outcome of interest. For example, financial literacy has a significant positive effect on individual trust. On the other hand, fsQCA revealed two different combinations of these antecedents that lead to high levels of trust and three different combinations that lead to low levels of trust. Although linear modeling has its own relevance, our findings indicate that this conventional statistical approach does not capture the complexities of the real phenomenon.

5. Conclusion

Using questionnaire data collected in Italy during the second wave of the COVID-19 pandemic, this paper investigated the impact of the pandemic on individuals' trust in the banking system. Through an fsQCA approach, we revealed the existence of a joint causality relationship between financial literacy, individual perceptions of the COVID-19 pandemic, and high/low trust in financial institutions. Although the profiles of respondents vary, perception of a possible conspiracy played an important role in the level of trust in banks during the COVID-19 pandemic. We further demonstrated that financial literacy is a necessary but non-sufficient condition for achieving high trust. According to the evidence provided by Guiso et al. (2008), studies on the relationship between trust and the financial system are highly important for financial market participants to better understand the factors that contribute to determining the level of confidence in the banking system during a non-financial crisis.

One of the limitations of this research is the methodology relying on cross-sectional data. In particular, the nature of the data does not allow us to identify the presence of causal factors. However, the use of data derived from cross-sectional data was suitable given the effect of financial literacy on trust.

The research results can contribute to a better understanding of the conditions that establish trust between the banking system and the depositors/investors who have different degrees of relationship with the market. Traders should reflect on why several respondents do not trust the banking system and why they believe there is a conspiracy in place. Practitioners should try to identify the possible factors that lead respondents to assume that there is a conspiracy to mitigate the risks of a health crisis turning into a financial crisis. Based on the results, financial institutions should develop strategies to strengthen the level of confidence in financial markets and improve the relationship between market players and their clients.

CRedit authorship contribution statement

Aymen Ammari: Methodology. **Evita Allodi:** Formal analysis, Resources, Validation, Writing –original draft. **Dario Salerno:** Conceptualization, Investigation, Project administration, Supervision, Writing – review & editing. **Gian Paolo Stella:** Conceptualization, Data curation, Project administration, Software, Visualization, Writing –original draft.

Data Availability

The data that has been used is confidential.

Appendix A

Investing in Financial Markets (Van Rooij, Lusardi and Alessie, 2011)

Have you ever invested in Financial Markets?

a. Yes.

Table 5
Linear regression.

TRUST	Coef.	St.Err.	t-value	p-value	[95 % Conf	Interval]	Sig
FL	0.102	0.035	2.87	0.004	0.032	0.172	***
PPC	-.145	0.026	-5.67	0	-.196	-0.095	***
PC	-.083	0.026	-3.20	0.001	-.134	-0.032	***
PCM	0.341	0.026	13.08	0	0.29	0.392	***
CW	0.017	0.025	0.67	0.503	-.032	0.065	
FMI	-.022	0.017	-1.29	0.197	-.056	0.011	
INC	0.116	0.03	3.93	0	0.058	0.174	***
Constant	0.341	0.028	12.30	0	0.287	0.395	***
R-squared	0.220		Number of obs		1079		
F-test	43.065		Prob > F		0.000		
Akaike crit. (AIC)	-98.987		Bayesian crit. (BIC)		-59.116		

*** p < .01, ** p < .05, * p < .1.

b. No.

Financial Literacy (Klapper and Lusardi, 2019). The questions are presented in a multiple-choice format. **The correct answer is in bold.**

Suppose you have some money. Is it safer to put your money into one business or investment, or to put your money into multiple businesses or investments?

- a. One business or investment **b. Multiple businesses or investments.**
- c. Don't know d. Refused to answer.

Suppose over the next 10 years the prices of the things you buy double. If your income also doubles, will you be able to buy less than you can buy today, the same as you can buy today, or more than you can buy today?

- a. Less **b. The same** c. More d. Don't know e. Refused to answer.

Suppose you need to borrow \$100. Which is the lower amount to pay back: \$105 or \$100 plus 3 %?

- a. \$105 **b. \$100 plus 3 %** c. Don't know d. Refused to answer.

Suppose you put money in the bank for 2 years and the bank agrees to add 15 % per year to your account. Will the bank add more money to your account the second year than it did the first year, or will it add the same amount of money both years?

- a. More** b. The same.
- c. Don't know.
- d. Refused to answer.

Suppose you had \$100 in a savings account and the bank adds 10 % per year to the account. How much money would you have in the account after 5 years if you did not remove any money from the account?

- a. More than \$150.**
- b. Exactly \$150.
- c. Less than \$150.
- d. Don't know.
- e. Refused to answer.

Trust on Banking System (Sapienza e Zingales, 2012)

On a scale ranging from 1 = I do not trust at all to 7 = I Trust Completely, How much trust do you have in the Banking System?

Perceived profit gained from COVID-19 (ppc): On a scale ranging from Not at all = 1 to Very much = 7, Do you think the Banks had any gains during the pandemic?

Perceived conspiracy (pc)

On a scale of Not at All = 1 to Very Much = 7, do you think that COVID-19 is a creation intended by the financial markets to increase the profits of Banks?

Perceived effective COVID-19 management (pcm)

On a scale ranging from Not At All = 1 to Very Much = 7, how confident are you that the banking system can handle the COVID-19 challenge well?

Income (Adapted by National Financial Capability Study, 2018)

What is your approximate annual income, including wages, tips, investment income, public assistance, income from retirement plans, etc.? Would you say it is....

- a. Less than €10.000
- b. €10.000 - €20.000
- c. €20.000 - €40.000
- d. €40.000 - €80.000
- e. €80.000 or more
- f. Don't Know
- g. Prefer not to say

Impact of COVID-19 on Wealth

COVID-19 impact on wealth (cw): On a scale ranging from Strongly disagree = 1 to Completely agree = 7, how much do you agree with the following statement: I am concerned that the pandemic will have economic consequences.

References

- Alesina, A., La Ferrara, E., 2002. Who trusts others? *J. Public Econ.* 85 (2), 207–234.
- Ashraf, B.N., 2020. Stock markets' reaction to COVID-19: cases or fatalities? *Res. Int. Bus. Financ.* 54, 101249.
- Berardi, C., Antonini, M., Genie, M.G., et al., 2020. The COVID-19 pandemic in Italy: policy and technology impact on health and non-health outcomes. *Health Policy Technol.* 9 (4), 454–487. <https://doi.org/10.1016/j.hlpt.2020.08.019>.
- Bertin, P., Nera, K., Delouvée, S., 2020. Conspiracy beliefs, rejection of vaccination, and support for hydroxychloroquine: a conceptual replication-extension in the COVID-19 pandemic context. *Front. Psychol.* 11. <https://doi.org/10.3389/fpsyg.2020.565128>.
- Billisbekov, N., Sarfo, C., Dandis, A.O., Eid, M.A.H., 2021. Linking bank advertising to customer attitudes: the role of cognitive and affective trust. *Manag. Sci. Lett.* 11, 1083–1092. <https://doi.org/10.5267/j.msl.2020.11.031>.
- Bjornskov, C., 2007. Determinants of generalized trust: a cross-country comparison. *Public Choice* 130, 1–21. <https://doi.org/10.1007/s11127-006-9069-1>.
- Buriak, A., Vozňáková, I., Sulowska, J., Kryvyh, Y., 2019. Social trust and institutional (Bank) trust: Empirical evidence of interaction. *Econ. Sociol.* 12 (4), 116–129. <https://doi.org/10.14254/2071-789X.2019/12-4/7>.
- Cahyono, E.F., Rani, L.N., Mardianto, M.F.F., 2021. Determinants of Indonesian Conventional and Islamic Bank Depositor Trust During the Covid-19 Pandemic. *J. Islam. Monet. Econ. Financ.* 7 (1), 1–32. <https://doi.org/10.21098/jimf.v7i1.1352>.
- Calcagno, R., Monticone, C., 2015. Financial literacy and the demand for financial advice. *J. Bank. Financ.* 50, 363–380.
- Carbó-Valverde, S., Maquí-López, E., and Rodríguez-Fernandez, F., 2013. Trust in banks: evidence from the Spanish financial crisis. 26th Australasian finance and banking conference 2013.
- Crujisen, C., Haan, J., Roerink, R., 2021. Financial knowledge and trust in financial institutions. *J. Consum. Aff.* 55 (2), 680–714. <https://doi.org/10.1111/joca.12363>.
- Dattalo, P., 2008. Determining sample size: Balancing power, precision, and practicality. oxford university press.
- De Jager, C.E., 2017. A question of trust: the pursuit of consumer trust in the financial sector by means of EU legislation. *J. Consum. Policy* 40, 25–49. <https://doi.org/10.1007/s10603-016-9334-8>.
- Devos, T., Spini, D., Schwartz, S.H., 2002. Conflicts among human values and trust in organizations. *Br. J. Soc. Psychol.* 41 (4), 481–494.
- Dijk, O., 2017. Bank run psychology. *J. Econ. Behav. Organ.* 144, 87–96. <https://doi.org/10.1016/j.jebo.2017.08.005>.
- Dufwenberg, M., 2015. Banking on experiments? *J. Econ. Stud.* 42 (6), 943–971. <https://doi.org/10.1108/JES-08-2015-0145>.
- Ennew, C., Sekhon, H., 2007. Measuring trust in financial services: the Trust Index. *Consum. Policy Rev.* 17 (2), 62–68. (<http://www.which.net>).
- Ermisch, J., Gambetta, D., Laurie, H., Siedler, T., Uhrih, S.C.N., 2009. Measuring people's trust. *J. R. Stat. Soc. Ser. A* 172 (Part 4), 749–769. <https://doi.org/10.1111/j.1467-985X.2009.00591.x>.
- Fisch, J.E., Seligman, J.S., 2022. Trust, financial literacy, and financial market participation. *J. Pension Econ. Financ.* 21 (4), 634–664.
- Flesia, L., Fietta, V., Colicino, E., Segatto, B., Monaro, M., 2020. Stable psychological traits predict perceived stress related to the COVID-19 outbreak. *PsyArXiv*. <https://doi.org/10.31234/osf.io/yb2h8>.
- Fungáčová, Z., Hasan, I., Weill, L., 2019. Trust in banks. *J. Econ. Behav. Organ.* 157, 452–476. <https://doi.org/10.16/j.jebo.2017.08.014>.
- Fungáčová, Z., Kerola, E., Weill, L., 2022. Does experience of banking crises affect trust in banks? *J. Financ. Serv. Res.* 62, 61–90. <https://doi.org/10.1007/s10693-021-00365-w>.
- Furnham, A., 2013. Commercial conspiracy theories: a pilot study. *Front. Psychol.* 4. <https://doi.org/10.3389/fpsyg.2013.00379>.
- Gaspar, R.M., Henriques, P.L., Corrente, A.R., 2020. Trust in Financial Markets: the Role of the Human Element. *Rev. Bras. De. Gest. De. Neg.* <https://doi.org/10.7819/rbgn.v22i3.4072>.
- Georgiou, N., Delfabbro, P., Balzan, R., 2020. COVID-19-related conspiracy beliefs and their relationship with perceived stress and pre-existing conspiracy beliefs. *Personal. Individ. Differ.* 166. <https://doi.org/10.1016/j.paid.2020.110201>.
- Giuliano, P., Spilimbergo, A., 2014. Growing Up in a Recession. *Rev. Econ. Stud.* 81, 787–817. <https://doi.org/10.1093/restud/rdt040>.
- Goedde-Menke, M., Langer, T., Pfingsten, A., 2014. Impact of the financial crisis on bank run risk – Danger of the days after. *J. Bank. Financ.* 40, 522–533. <https://doi.org/10.1016/j.jbankfin.2013.11.028>.
- Grayson, K., Johnson, D., Chen, D.F.R., 2008. Is firm trust essential in a trusted environment? How trust in the business context influences customers. *Journal of Marketing Research* 45 (2), 241–256.
- Guiso, L. (2010). A trust-driven financial crisis. Implications for the future of financial markets. Einaudi Institute for Economics and Finance, Working Paper 1006. Available at: <http://ideas.repec.org/p/eie/wpaper/1006.html>.
- Guiso, L., Sapienza, P., Zingales, L., 2008. Trusting the stock market. *J. Financ.* 63 (6), 2557–2600.
- Hansen, T., 2012. Understanding trust in financial services: the influence of financial healthiness, knowledge, and satisfaction. *J. Serv. Res.* 15 (3), 280–295.
- Hansen, T., 2014. The role of trust in financial customer–seller relationships before and after the financial crisis. *J. Consum. Behav.* 13 (6), 442–452.
- Harms, R., Kraus, S., & Reschke, C.H. (2007). Configurations of new ventures in entrepreneurship research: contributions and research gaps. *Management Research News*.
- Hasan, I., Politidis P.N. and Sharma, Z., 2021. Global syndicated lending during the COVID-19 pandemic, *Journal of Banking and Finance*, <https://doi.org/10.1016/j.jbankfin.2021.106121>.
- Imhoff, R., Lamberty, P., Klein, O., 2018. Using power as a negative cue: How conspiracy mentality affects epistemic trust in sources of historical knowledge. *Personal. Soc. Psychol. Bull.* 44 (9), 1364–1379. <https://doi.org/10.1177/0146167218768779>.
- Jaffer, S., Morris, N., Vines, D., 2014. Why trustworthiness is important. In: Morris, N., Vines, D. (Eds.), *Capital Failure: Rebuilding Trust in Financial Services*. Oxford University Press, Oxford.
- Jansen, D., Mosch, R.H.J., Van der Crujisen, C., 2015. When does the general public lose trust in banks? *J. Financ. Serv. Res.* 48, 127–141. <https://doi.org/10.1007/s10693-014-0201-y>.
- Jarvinen, R.A., 2014. Consumer Trust in Banking Relationship in Europe. *Int. J. Bank Mark.* 32 (6), 551–566.
- John, K., Li, J., 2021. nCOVID-19, Volatility Dynamics, and Sentiment Trading. *J. Bank. Financ.* <https://doi.org/10.1016/j.jbankfin.2021.106162>.
- Jolley, D., Douglas, K.M., 2014. The social consequences of Conspiracy: Exposure to conspiracy theories decreases intentions to engage in politics and to reduce one's carbon footprint. *Br. J. Psychol.* 105 (1), 35–56. <https://doi.org/10.1111/bjop.12018>.
- Kersting, L.M., Marley, R.N., Mellon, M.J., 2015. The association between financial literacy and trust in financial markets among novice nonprofessional investors. *Acad. Account. Financ. Stud. J.* 19 (3), 201.
- Klapper, L., Lusardi, A., 2019. Financial literacy and financial resilience: Evidence from around the world. *Financ. Manag.* 49 (3), 589–614. <https://doi.org/10.1111/fima.12283>.
- Knack, S., Keefer, P., 1997. Does social capital have an economic payoff? A cross-country investigation. *Q. J. Econ.* 112 (4), 1251–1288.
- Knell, M., Stix, H., 2015. Trust in Banks during Normal and Crisis Times—Evidence from Survey Data. *Economica* 82 (1), 995–1020. <https://doi.org/10.1111/ecca.12162>.
- Lerner, J.S., Keltner, D., 2001. Fear, anger, and risk. *J. Personal. Soc. Psychol.* 81, 146–159.
- Lerner, J.S., Small, D.A., Loewenstein, G., 2004. Heart strings and purse strings: Carryover effects of emotions on economic decisions. *Psychol. Sci.* 15, 337–341.
- Li, X., Feng, H., Zhao, S., Carter, D.A., 2021. The effect of revenue diversification on bank profitability and risk during the COVID-19 pandemic. *Financ. Res. Lett.* 43. <https://doi.org/10.1016/j.frl.2021.101957>.
- Lin, W.R., Wang, Y.H., Hung, Y.M., 2020. Analyzing the factors influencing adoption intention of internet banking: Applying DEMATEL-ANP-SEM approach. *PLoS ONE* 15 (2), 1–25. <https://doi.org/10.1371/journal.pone.0227852>.
- Liu, N., Zhang, F., Wei, C., Jia, Y., Shang, Z., Sun, L., et al., 2020. Prevalence and predictors of PTSS during COVID-19 outbreak in China hardest-hit areas: gender differences matter. *Psychiatry Res.* 287. <https://doi.org/10.1016/j.psychres.2020.112921>.

- Liu, Z., Huynh, T.L.D., Dai, P.F., 2021. The impact of COVID-19 on the stock market crash risk in China. *Res. Int. Bus. Financ.* 57, 101419.
- Losada-Baltar, A., Jiménez-Gonzalo, L., Gallego-Alberto, L., Pedrosa-Chaparro, M.D.S., Fernandes-Pires, J., Márquez-González, M., 2020. We're staying at home. Association of self-perceptions of ageing, personal and family resources and loneliness with psychological distress during the lock-down period of COVID-19. *J. Gerontol. Ser. B: Psychol. Sci. Soc. Sci.* <https://doi.org/10.1093/geronb/25Fgbaa048>.
- Lusardi, A., Mitchell, O.S., 2014. The Economic Importance of Financial Literacy: Theory and Evidence. *J. Econ. Lit., Am. Econ. Assoc.* 52 (1), 5–44. <https://doi.org/10.1257/jel.52.1.5>.
- Malmendier, U., Nagel, S., 2016. Learning from Inflation Experiences. *Q. J. Econ.* 131 (1), 53–87. <https://doi.org/10.1093/qje/qjv037>.
- Mayer, R.C., Davis, J.H., Schoorman, F.D., 1995. An integrative model of organizational trust. *Acad. Manag. Rev.* 20 (3), 709–734.
- Nooteboom, B., 2010. Vertrouwen, besluitvorming en communicatie [Trust, decision-making, and communication]. *M&O* 6, 25–38.
- Núñez Letamendia, L., & Pohrer, B. (2020). The effect of financial literacy on trust: do financially literate individuals have more trust in the financial system. Working Paper, April.
- Omorieg, O.K., Addae, J.A., Coffie, S., Ampong, G.O.A., Ofori, K.S., 2019. Factors influencing consumer loyalty: evidence from the Ghanaian retail banking industry. *Int. J. Bank Mark.* 37 (3), 798–820. <https://doi.org/10.1108/IJBM-04-2018-0099>.
- Orrù, G., Ciachini, R., Gemignani, A., Conversano, C., 2020. Psychological intervention measures during the COVID-19 pandemic. *Clin. Neuropsychiatry* 17, 76–79.
- Pakpour, A.H., Griffiths, M.D., 2020. The fear of COVID-19 and its role in preventive behaviours. *J. Concurr. Disord.* 2 (1), 58–63. ISSN 2562-7546.
- Patel, R., Migliavacca, M., Oriani, M., 2022. Blockchain in banking and finance: is the best yet to come? A bibliometric review. *Res. Int. Bus. Financ.*, 101718.
- Polizzi, C., Lynn, S.J., Perry, A., 2020. Stress and coping in the time of Covid-19: pathways to resilience and recovery. *Clin. Neuropsychiatry* 17, 59–62.
- Ragin, C.C., 2000. *Fuzzy-set Social Science*. University of Chicago Press, Chicago.
- Ragin, C.C., 2008. Measurement versus calibration: A set-theoretic approach. In *The Oxford handbook of political methodology*.
- Ragin, C.C., 2009. *Redesigning social inquiry: Fuzzy sets and beyond*. University of Chicago Press.
- Rihoux, B., Ragin, C.C., 2008. *Configurational Comparative Methods: Qualitative Comparative Analysis (QCA) and Related Techniques*, Vol. 51. Sage Publications, Los Angeles.
- Rousseau, D.M., Sitkin, S.B., Burt, R.S., Camerer, C., 1998. Not SO Different after All: A Cross-discipline View Of Trust. *Acad. Manag. Rev.* 23 (3) <https://doi.org/10.5465/amr.1998.926617>.
- Sapienza, P., Zingales, L., 2012. A trust crisis. *Int. Rev. Financ.* 12 (2), 123–131. <https://doi.org/10.1111/j.1468-2443.2012.01152.x>.
- Sharma, S.K., 2019. Integrating cognitive antecedents into TAM to explain mobile banking behavioral intention: A SEM-neural network modeling. *Inf. Syst. Front.* 21 (4), 815–827. <https://doi.org/10.1007/s10796-017-9775-x>.
- Shim, S., Serido, J., Tang, C., 2013. After the global financial crash: individual factors differentiating young adult consumers' trust in banks and financial institutions. *J. Retail. Consum. Serv.* 20 (1), 26–33.
- Uslaner, E.M., 2018. *The Oxford Handbook of Social and Political Trust*. Oxford University Press.
- van Der Cruysen, C., de Haan, J., Roerink, R., 2021a. Financial knowledge and trust in financial institutions. *J. Consum. Aff.* 55 (2), 680–714.
- Van der Cruysen, C., Jonker, N., 2019. Pension profile preferences: the influence of trust and expected expenses. *Appl. Econ.* 51 (12), 1212–1231.
- Van der Cruysen, C., de Haan, J., Roerink, R., 2021b. Trust in financial institutions: a survey. *J. Econ. Surv.* 1–41. <https://doi.org/10.1111/joes.12468>.
- Van der Cruysen, C., de Haan, J., Jonker, N., 2022. Has the COVID-19 pandemic affected public trust? Evidence for the US and the Netherlands. *J. Econ. Behav. Organ.* 200, 1010–1024. <https://doi.org/10.1016/j.jebo.2022.07.006>.
- Van Prooijen, J.W., Acker, M., 2015. The influence of control on belief in conspiracy theories: conceptual and applied extensions. *Appl. Cogn. Psychol.* 29, 753–761.
- Van Rooij, M., Lusardi, A., Alessie, R., 2011. Financial literacy and stock market participation. *J. Financ. Econ.* 101 (2), 449–472. <https://doi.org/10.1016/j.jfineco.2011.03.006>.
- Wagemanna, C., Schneider, C.Q., 2010. Qualitative comparative analysis (QCA) and fuzzy-sets: agenda for a research approach and a data analysis technique. *Comp. Sociol.* 9, 1–21.
- Wälti, S., 2012. Trust no more? The impact of the crisis on citizens' trust in central banks. *J. Int. Money Financ.* 31, 593–605. <https://doi.org/10.1016/j.jimonfin.2011.11.012>.
- Wang, C., Pan, R., Wan, X., Tan, Y., Xu, L., Ho, C.S., et al., 2020. Immediate psychological responses and associated factors during the initial stage of the 2019 coronavirus disease (COVID-19) epidemic among the general population in China. *Int. J. Environ. Res. Public Health* 17 (5). <https://doi.org/10.3390/ijerph17051729>.